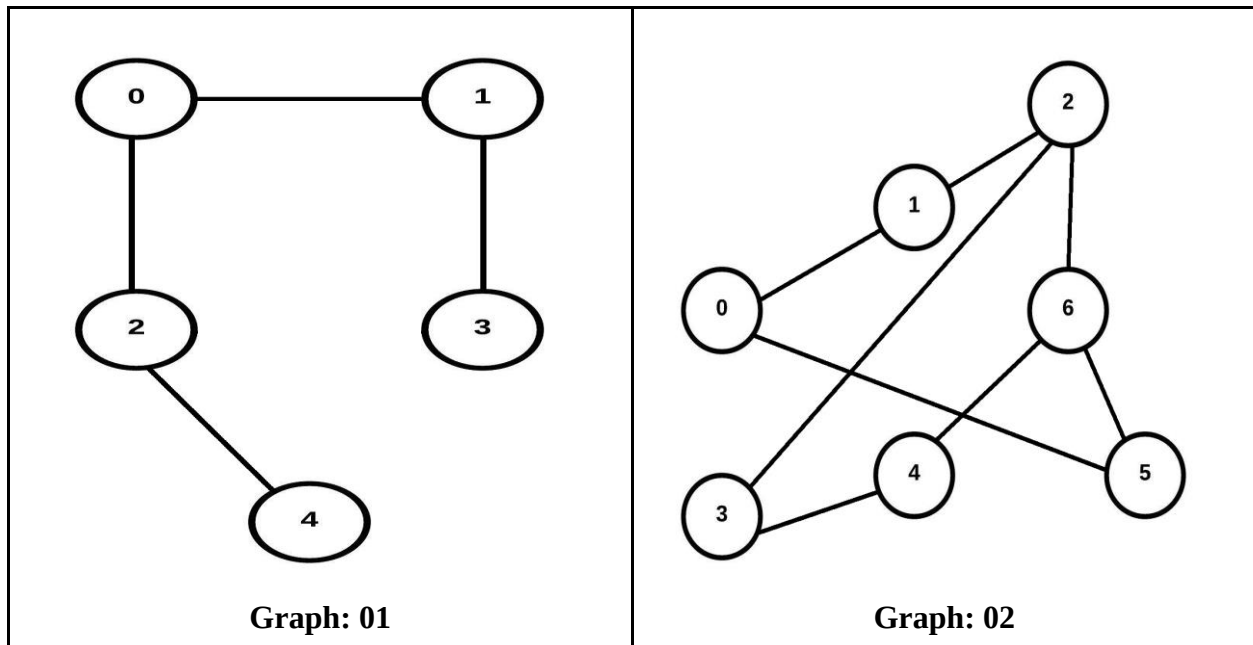


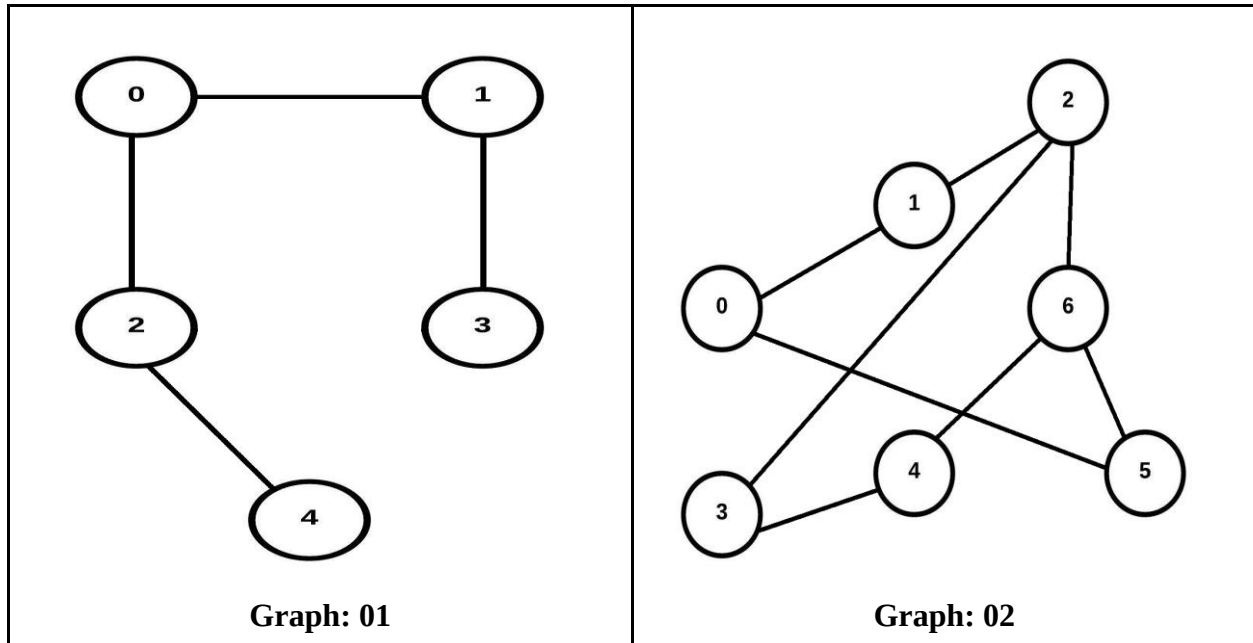
Question 01:



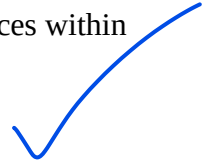
You are given a graph G with N vertices. Implement a function to find the shortest path (minimum number of edges) between two given vertices u and v using **Breadth-First Search**.

Sample Input	Sample Output
for Graph 01: Source: 0, Destination: 3	2. <i>Explanation: Path from 0 to 3 has 2 edges. (0-1-3)</i>
For Graph 02: Source: 1, Destination: 5	2. <i>Explanation: There are three path from 0 to 5. One is (1-2-3-4-6-5), another is (1-2-6-5) and last one is (1-0-5). Among these, shortest path is (1-0-5) and it has two edges.</i>

Question 02:

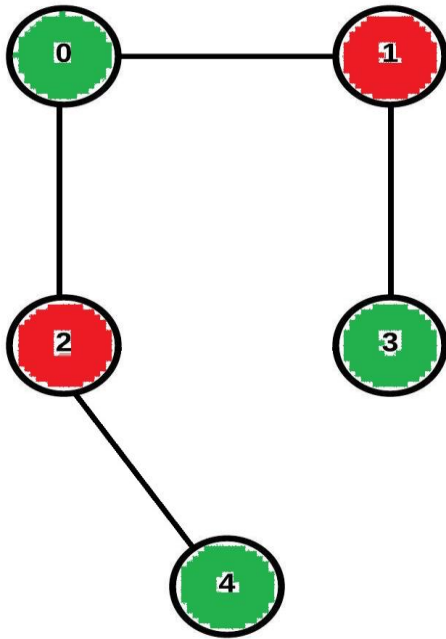


Write a program to check if graph G is bipartite using Depth-first Search. A bipartite graph is one in which the vertices can be divided into two disjoint sets such that no two vertices within the same set are adjacent



Sample Input	Sample Output
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Graph 01

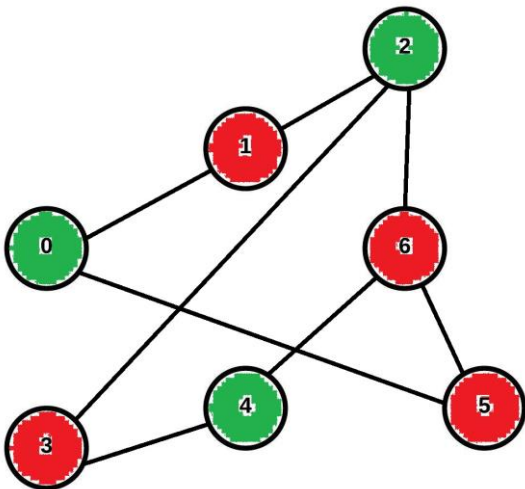


TRUE.

Explanation: No two vertices that are connected together has same color.

N.B. You don't have to show the graph as output. Output needs to print only "true" or "false".

Graph 02

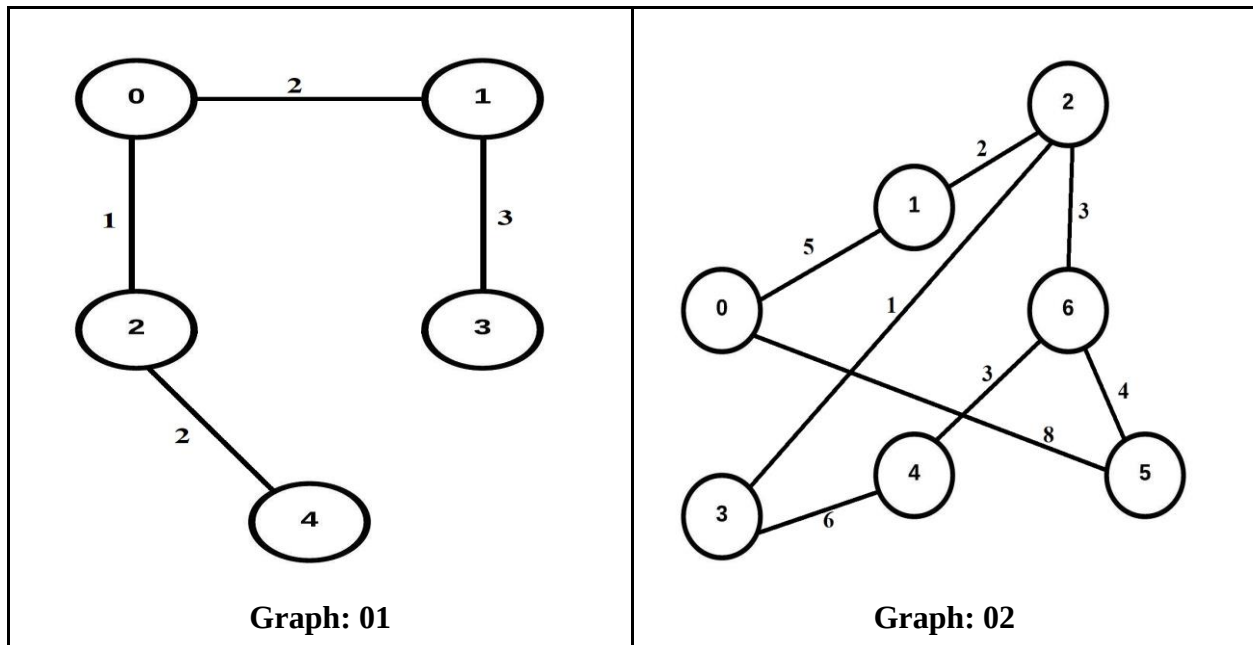


FALSE.

Explanation: Vertices 5 and 6 are connected together and has same color.

N.B. You don't have to show the graph as output. Output needs to print only "true" or "false".

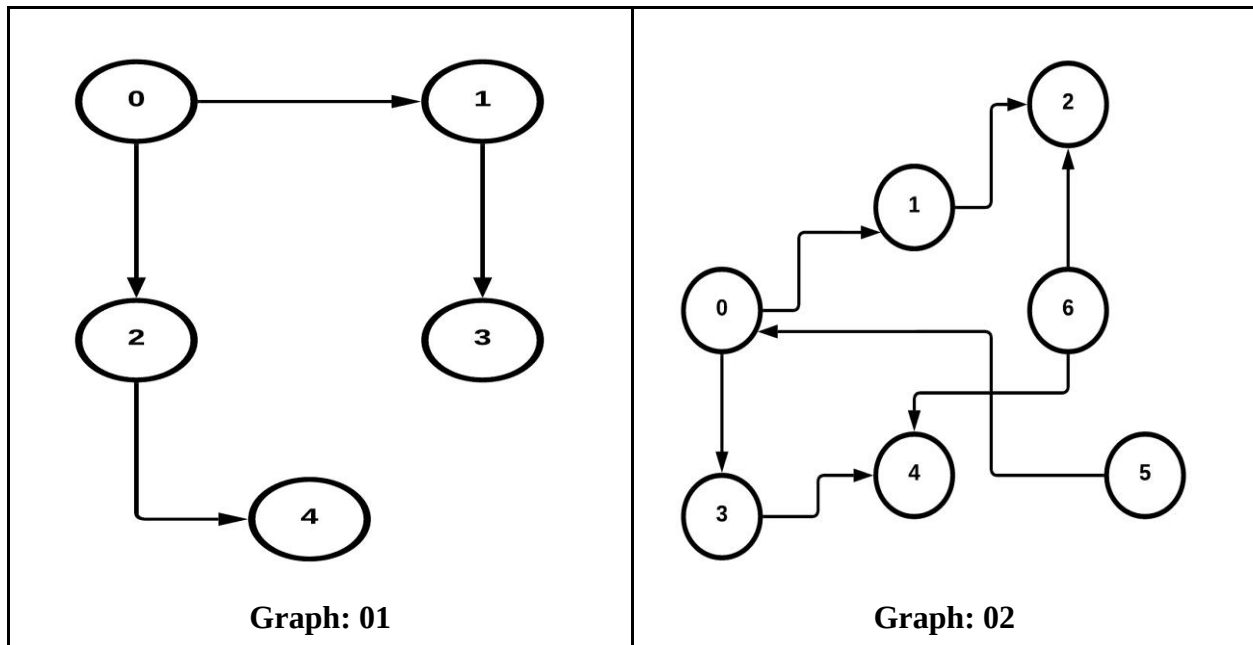
Question 03:



Two weighted graph is given. You need to find the edge with maximum weight between the source and destination nodes.

Sample Input	Sample Output
for Graph 01: Source: 0, Destination: 3	3. <i>Explanation: Path from 0 to 3 has 2 edges. (0-1-3). Weight of 0-1 is 2. And weight of 1-3 is 3. Among these 2 weighted edges, 3 is max.</i>
For Graph 02: Source: 1, Destination: 5	8. <i>Explanation: The shortest path from 1 to 5 is (1-0-5). And weight of these two edges are 5 and 8. And 8 is max here.</i>

Question 04:



From a directed acyclic graph, you have to find if two nodes have a path or not, using Depth-First search.

Sample Input	Sample Output
for Graph 01: Source: 0, Destination: 3	true. <i>Explanation: Path from 0 to 3 exists. (0-1-3)</i>
For Graph 02: Source: 1, Destination: 5	false. <i>Explanation: path from 1 to 5 does not exist. (1-2). From 2, there is no outgoing edge.</i>